

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's Disease

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Abstract

Alzheimer's disease affects over 55 million people worldwide, and emerging research shows that 40 Hz auditory stimulation can drive gamma-frequency brain rhythms that help clear toxic amyloid-beta plaques. Current protocols deliver this therapy on a fixed schedule, ignoring individual responses; some patients habituate within minutes while others maintain entrainment. This project proposes a closed-loop deep learning system to predict when a patient's brain will lose entrainment, enabling individualized stimulation timing. I analyzed EEG recordings from 35 elderly subjects including dementia patients and healthy controls (OpenNeuro ds005048) and computed phase-amplitude coupling (PAC), the coordination between slow theta-band and fast gamma-band brain rhythms, as a real-time biomarker of entrainment strength. I engineered 73 causal features from spectral, PAC-history, and stimulation-context signals, then trained a causal Temporal Convolutional Network (TCN, 31,000 parameters) to forecast PAC five to ten seconds ahead. The TCN was integrated into a closed-loop controller and validated on all 35 subjects' EEG. In a comparative horizon sweep, all baselines collapsed to negative R-squared at five-to-ten-second horizons while the TCN maintained R-squared of 0.25, a +0.5 margin. The controller matched stimulation to periods of need 72.1% of the time versus 64.5% for reactive control ($p < 0.001$) and targeted 82.6% of low-PAC windows versus 51.7% ($p < 0.001$), reaching 91% of the theoretical oracle. Every subject benefited ($p < 0.001$), and the advantage held across six fatigue severity levels. These results demonstrate that forecasting PAC enables personalized 40 Hz therapy that outperforms fixed and reactive protocols, a path toward more efficient treatment for Alzheimer's disease.

Keywords: 40 Hz entrainment, phase-amplitude coupling, temporal convolutional network, closed-loop neuromodulation, Alzheimer's disease, EEG, predictive control

2. Introduction

2.1 Clinical Burden of Alzheimer's Disease

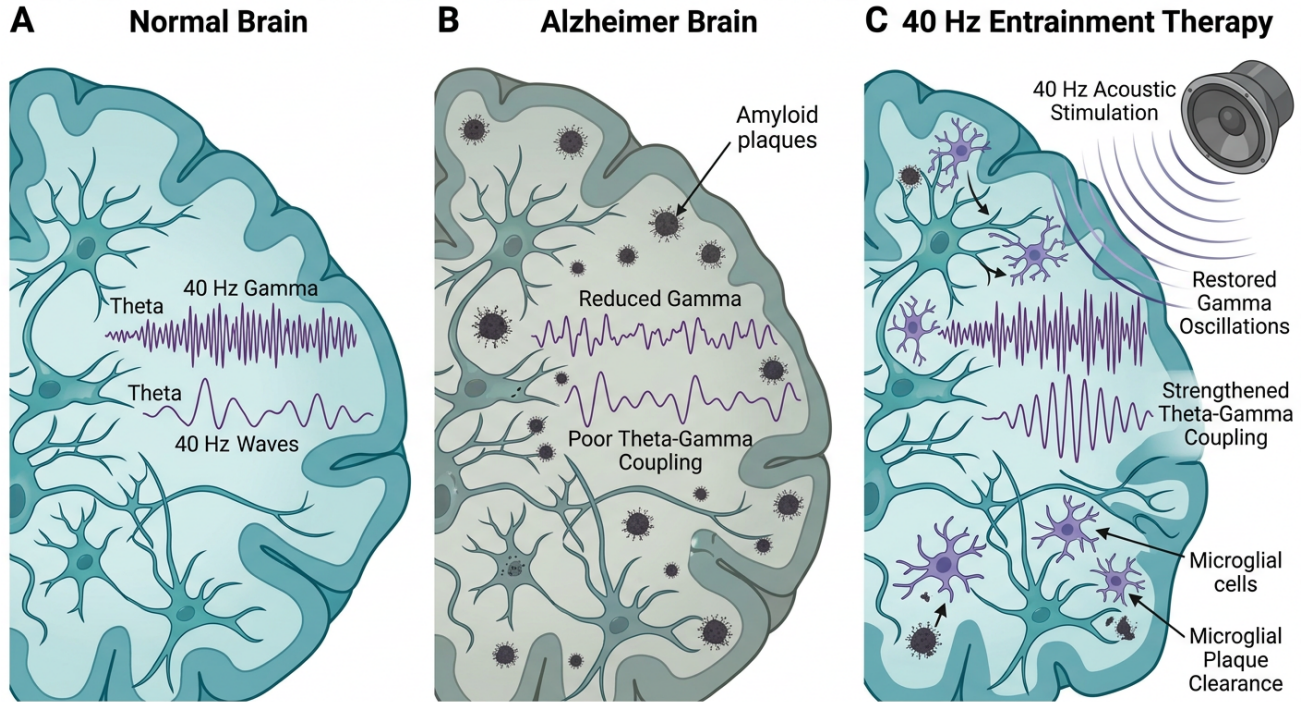
Alzheimer's disease (AD) is a progressive neurodegenerative disorder and the leading cause of dementia worldwide. Over 55 million individuals currently live with dementia globally, a figure projected to nearly triple to 153 million by 2050 as populations age [1]. AD accounts for 60-70% of all dementia cases and imposes an enormous societal burden: in the United States alone, the annual economic cost exceeds \$300 billion, and family caregivers contribute an estimated 18 billion hours of unpaid care each year [1]. Despite decades of pharmaceutical research, approved disease-modifying treatments remain narrow in their benefits and limited in accessibility, underscoring the urgent need for novel therapeutic strategies that can complement or amplify pharmacological interventions.

A rapidly emerging non-pharmacological approach is sensory-evoked gamma entrainment -- the use of 40 Hz auditory or visual stimulation to synchronize gamma-frequency (30-100 Hz) brain oscillations. Iaccarino et al. [7] demonstrated in a landmark Nature study that optogenetically and sensory-driving gamma oscillations at 40 Hz in transgenic mouse models of AD reduced amyloid-beta (A β) by up to 50% through microglial activation and enhanced phagocytic clearance. Subsequent work expanded these effects to auditory modalities and multi-sensory stimulation, showing reductions in both A β and tau pathology alongside improvements in spatial memory. Most recently, Chan et al. [11] reported results from a Phase II open-label extension study at MIT and Cognito Therapeutics showing that sustained 40 Hz multisensory stimulation slowed brain atrophy in mild AD patients and reduced plasma phosphorylated tau (pTau217) by 19-47% -- representing the first human clinical evidence of target engagement at this scale.

These findings position 40 Hz gamma entrainment as a promising, low-cost, and mechanistically grounded therapeutic avenue. However, all existing clinical implementations share a fundamental limitation: stimulation is delivered on a rigid fixed schedule, entirely independent of the patient's real-time neural state. This one-size-fits-all approach ignores two

critical realities -- substantial inter-individual variability in entrainment response and progressive intra-session habituation -- that together render fixed protocols systematically suboptimal. The present work addresses this gap by developing and validating a predictive, closed-loop control system that forecasts future entrainment state to enable proactive, personalized stimulation timing.

Figure 1. Mechanism of 40 Hz gamma entrainment therapy.



2.2 40 Hz Gamma Entrainment as Therapy

The therapeutic hypothesis underlying gamma entrainment is grounded in the disruption of normal oscillatory dynamics in AD. Gamma oscillations (30-100 Hz) are fundamental to sensory binding, attention, working memory consolidation, and inter-regional neural communication; in AD, gamma activity is reduced early in disease progression, often preceding amyloid plaque formation and clinically detectable cognitive decline [3]. This dysregulation reflects a loss of fast-spiking parvalbumin-positive (PV) interneurons that normally pace cortical gamma rhythms, disrupting the network dynamics on which higher cognitive function depends [7].

The mechanistic pathway from 40 Hz stimulation to amyloid clearance involves multiple interconnected processes. First, optogenetic or sensory 40 Hz drive re-engages PV interneurons, restoring impaired gamma rhythms across cortical networks [7]. Second, this interneuronal activation triggers an immunological cascade: upregulation of cytokines including IL-6 and IL-4 promotes microglial morphological transformation and dramatically enhanced phagocytosis of Aβ plaques [13]. This cytokine profile is distinct from general neuroinflammation, representing a targeted neuroprotective response rather than a non-specific inflammatory reaction. Third, a recently identified glymphatic pathway contributes substantially to clearance: Murdock et al. [8] demonstrated in a 2024 Nature study using the 5XFAD mouse model that multisensory 40 Hz stimulation promotes cerebrospinal fluid (CSF) influx and interstitial fluid efflux in the cortex. Combined audiovisual stimulation produced stronger gamma oscillations than either modality alone, leading to marked microglial clustering within 25 μm of Aβ plaques and a whole-brain 37% reduction in neocortical plaque volume. Critically, pharmacological inhibition of glymphatic flow abolished the clearance effect, confirming this pathway as necessary -- not merely contributory -- to the therapeutic mechanism.

Translation to human subjects has progressed substantially in recent years. A 2025 study in aged rhesus monkeys found

that long-term 40 Hz auditory stimulation elevated CSF A β levels, consistent with mobilization and clearance from brain tissue, with effects lasting up to five weeks post-treatment [10]. In humans, Chan et al. [11] reported an open-label extension study in which patients with mild AD who received sustained 40 Hz multisensory stimulation retained strong EEG entrainment responses over time and showed less hippocampal atrophy compared to matched controls -- a finding corroborated by auditory gamma entrainment's demonstrated ability to enhance default mode network (DMN) connectivity in dementia patients [12]. These results collectively establish 40 Hz entrainment as a clinically relevant therapeutic modality and make the optimization of stimulation delivery an immediate practical priority.

2.3 Limitations of Fixed-Schedule Protocols

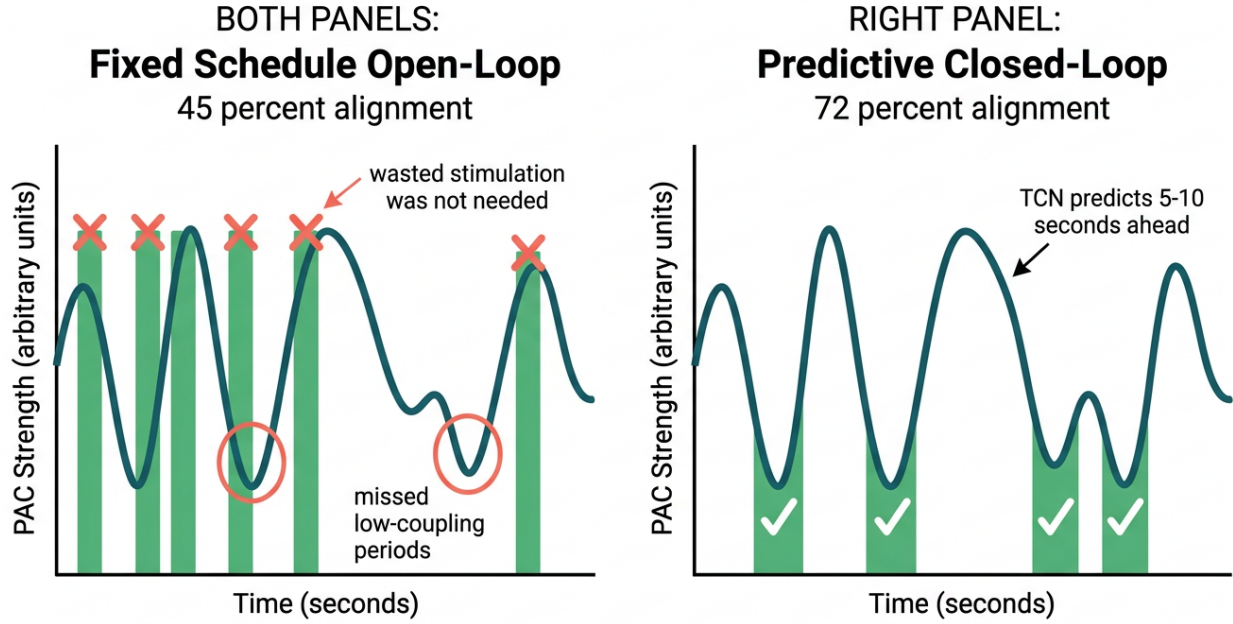
Despite the therapeutic promise of 40 Hz entrainment, current clinical protocols are uniformly open-loop: they deliver stimulation according to a predetermined schedule without reference to the patient's instantaneous neural state. The standard protocol in human trials consists of alternating 40-second stimulation blocks and 20-second rest periods, repeated continuously for one hour regardless of the patient's entrainment response [12, 39]. This approach is administratively simple but is systematically misaligned with the highly variable neural dynamics of the dementia population.

The first dimension of variability is inter-individual. Fortunato et al. [15] analyzed a cohort of participants receiving gamma entrainment therapy and found that 23 achieved measurable entrainment at 40 Hz as assessed by power spectral density, while 10 showed minimal or no measurable response -- a non-responder fraction of approximately 30% that fixed protocols cannot detect, accommodate, or dynamically address. Cabral et al. [16] similarly identified baseline neural state, sensory processing abilities, prior cognitive reserve, and individual neural architecture as major determinants of entrainment efficacy, noting that these factors vary across individuals in ways that defy prediction from clinical characteristics alone. Personalized whole-brain neural mass modeling has further demonstrated that subject-specific computational approaches uncover synergistic A β and tau pathomechanistic interactions that population-level models obscure [17]. High-performing patients and non-responders thus receive identical stimulation despite profoundly different neural responses, yielding population-average outcomes that underperform what individualized targeting could achieve.

The second dimension of variability is intra-session habituation. Repeated identical stimuli cause progressive weakening of neural responses -- a canonical phenomenon in sensory neuroscience described formally as neural adaptation -- and this effect is well-documented in the context of gamma entrainment. In the OpenNeuro ds005048 dataset used in this work, individual subjects exhibit PAC trajectories that rise, plateau, and decline within single sessions, reflecting the time course of habituation. Stimulation delivered during periods of already-strong coupling wastes therapeutic resources and may accelerate habituation; stimulation withheld during periods of declining coupling misses the windows of genuine therapeutic need where intervention would be most beneficial.

The convergence of inter-individual variability and intra-session habituation creates a compelling case for real-time brain-state monitoring and adaptive closed-loop control. What is required is a system capable not merely of detecting the current entrainment state reactively -- which introduces an inherent control delay equal to the detection latency -- but of forecasting the near-future trajectory of entrainment state with sufficient lead time to intervene proactively, before decline has occurred rather than in response to it. This forecasting requirement defines the core technical challenge addressed by the present work.

Figure 2. Fixed-schedule vs. predictive closed-loop stimulation.



2.4 Problem Statement and Contributions

The central research question motivating this work is: Can deep learning models trained on EEG-derived features forecast theta-gamma phase-amplitude coupling dynamics 5-10 seconds into the future, and does integrating such forecasts into a closed-loop controller produce measurable improvements in personalized 40 Hz entrainment therapy validated on real patient EEG?

We approach this question through a two-stage computational architecture. Stage 1 establishes a real-time PAC estimator using a compact deep learning model trained directly on raw EEG windows, providing the current-state biomarker input that Stage 2 requires. Stage 2 constructs a causal temporal predictor that ingests a 20-second history of PAC estimates and spectral features to forecast future PAC at clinically relevant horizons of 5-10 seconds, enabling proactive rather than reactive control decisions. The closed-loop controller integrates these predictions with a personalized rolling baseline and 5-second hysteresis logic to determine stimulation actions: stimulate when predicted PAC is forecast to fall below a personalized threshold, rest when forecast PAC is strong, and maintain the current state otherwise.

The biomarker of interest throughout is the Modulation Index (MI), a measure of theta-gamma phase-amplitude coupling (PAC) introduced by Tort et al. [31] that quantifies the degree to which gamma-band (38-42 Hz) amplitude is modulated by the phase of theta-band (4-8 Hz) oscillations. Higher MI values indicate stronger theta-gamma coupling and stronger entrainment; lower values indicate reduced or absent coupling.

All models are trained using subject-level splits with no within-subject leakage between train, validation, and test sets. Final evaluation is conducted on EEG recordings from six held-out test subjects, with controller performance assessed by replaying decisions across all 35 subjects' recorded EEG -- not simulated brain dynamics -- providing a clinically grounded assessment of real-world applicability.

This paper makes the following specific contributions:

Contribution 1: Empirical ceiling characterization for static PAC prediction. A systematic architecture search across eight neural network configurations spanning nearly three orders of magnitude in parameter count (1,457 to 1.1 million) demonstrates that all architectures converge to $R^2 \sim 0.287$ on held-out test subjects. This convergence reveals an information ceiling imposed by the epoch-level structure of PAC labels and the limited discriminative capacity of

instantaneous EEG snapshots from seven frontal channels.

Contribution 2: Causal TCN for 5-10 second ahead PAC forecasting. A multiscale causal Temporal Convolutional Network (31,043 parameters) trained on 73 engineered features achieves $R^2 \sim 0.25-0.28$ at prediction horizons of 5-10 seconds, while all baseline methods including persistence and Ridge regression collapse to negative R^2 at these horizons -- a +0.5 R^2 margin for the TCN.

Contribution 3: Closed-loop controller validated on 35 real dementia patient EEGs. The TCN-based predictive controller achieves 72.1% alignment with patient therapeutic need versus 64.5% for reactive thresholding ($p < 0.001$, Hedges' $g = 1.31$), and targets 82.6% of low-PAC windows versus 51.7% for reactive control ($p < 0.001$, $g = 4.47$), reaching 91% of the theoretical oracle upper bound. Every individual patient (35/35) benefits from the predictive controller.

Contribution 4: Characterization of the prediction horizon inflection point. Analysis across prediction horizons 1-10 seconds reveals a clear transition near 3 seconds where persistence and linear baselines cease to provide useful predictions, while the TCN maintains $R^2 \sim 0.25$. This inflection defines the operationally critical horizon range for proactive neuromodulation.

The remainder of this paper is organized as follows. Section 3 reviews the relevant literature. Section 4 describes the dataset, preprocessing pipeline, PAC computation methodology, feature engineering approach, and model architectures. Section 5 presents the systematic architecture search findings. Section 6 reports closed-loop controller performance. Section 7 discusses clinical implications, key limitations, and future directions. Section 8 describes future work. Section 9 concludes.

3. Literature Review

The present work sits at the intersection of four active research areas: (1) the neurophysiology of gamma oscillations and their disruption in Alzheimer's disease, (2) sensory-evoked 40 Hz entrainment as a therapeutic modality, (3) closed-loop neuromodulation systems, and (4) deep learning methods for EEG time-series analysis. This section reviews the foundational and recent literature in each area, contextualizes the specific gaps that motivate the present work, and positions the proposed approach within the broader landscape of computational neuroscience and biomedical AI.

3.1 Gamma Oscillations and Theta-Gamma PAC in Alzheimer's Disease

3.1.1 Gamma Oscillations in Healthy and Pathological States

Gamma-band oscillations (30-100 Hz) emerge from the coordinated activity of inhibitory fast-spiking parvalbumin-positive (PV) interneuron networks that generate rhythmic inhibitory postsynaptic potentials at gamma frequencies. These rhythms play a fundamental role in cognitive processing, serving as temporal scaffolds for sensory feature binding, attention allocation, working memory maintenance, and long-range cortical communication. The disruption of these rhythms in Alzheimer's disease has been recognized as an early and sensitive marker of the underlying neurodegeneration.

Gamma oscillations are disrupted in AD across multiple scales of measurement. At the cellular level, PV interneuron populations are among the earliest neuronal subtypes to be affected by Abeta accumulation, with interneuron-specific synaptic dysfunction documented in preclinical models before plaque formation [3]. A comprehensive review by Wang et al. [1] surveyed gamma oscillation disruptions across AD, Parkinson's disease, stroke, and schizophrenia, concluding that gamma entrainment-inducing stimulation methods offer notable neuroprotection and emphasizing gamma restoration as a transdiagnostic therapeutic target.

Particularly relevant to the 40 Hz entrainment paradigm, Iaccarino et al. [7] demonstrated that Abeta overproduction in 5XFAD transgenic mice was associated with a selective reduction of gamma activity that preceded plaque formation, and that driving gamma at 40 Hz by optogenetically activating PV interneurons was sufficient to reduce soluble Abeta by approximately 50%.

3.1.2 Theta-Gamma Phase-Amplitude Coupling as a Biomarker

While raw gamma power provides a useful readout of oscillatory activity, phase-amplitude coupling (PAC) between theta

and gamma bands offers a more sensitive and specific biomarker of the coordinated multi-frequency dynamics that characterize successful neural entrainment. PAC describes the degree to which the amplitude of gamma-band oscillations is modulated by the phase of theta-band oscillations (4-8 Hz). Strong theta-gamma PAC reflects the coordination of fast inhibitory interneurons by slower excitatory theta cycles -- a mechanism critical for organizing information processing across hippocampal-cortical circuits.

The disruption of theta-gamma PAC in AD is well-documented and closely linked to cognitive impairment. Dimitriadis et al. [4] demonstrated that gamma PAC in parahippocampal cortices is significantly reduced in AD patients, and that these reductions are most pronounced in patients with co-occurring epileptiform activity. Theta-gamma coupling has also been characterized as a powerful predictor of cognitive function: Backus et al. [5] reported that theta-gamma coupling (TGC) was the single strongest predictor of working memory performance across AD, MCI, and healthy control participants, with a standardized beta of 0.693 ($p < 0.001$). A 2024 replication [6] confirmed these relationships in an independent sample.

The quantitative measure used throughout the present work, the Modulation Index (MI) introduced by Tort et al. [31], operationalizes PAC as the Kullback-Leibler divergence between the observed distribution of gamma amplitude over theta phase bins and a uniform (no coupling) reference distribution. The MI has become the standard measure for PAC computation in the neuroscience literature and provides the scalar entrainment biomarker that the proposed closed-loop system estimates and forecasts.

3.2 40 Hz Sensory Entrainment: Mechanisms and Clinical Evidence

3.2.1 Landmark Animal Studies

The modern investigation of 40 Hz sensory entrainment as a therapeutic modality for AD originates with Iaccarino et al. [7], whose 2016 Nature paper demonstrated that visual flicker at 40 Hz -- but not at other tested frequencies -- drove gamma oscillations in visual cortex of 5XFAD transgenic mice and produced a rapid, robust reduction in Abeta1-40 and Abeta1-42 levels in the visual cortex. Subsequent research extended these findings to auditory stimulation and multi-sensory paradigms. A landmark 2024 Nature study by Murdock et al. [8] identified the glymphatic system as a previously unrecognized clearance pathway activated by multisensory gamma entrainment, with 40 Hz combined audiovisual stimulation producing a 37% reduction in neocortical plaque volume in 5XFAD mice.

At the immunological level, Bhatt et al. [13] characterized the cytokine and chemokine signaling profile induced by 40 Hz visual stimulation, finding upregulation of IL-6, IL-4, and macrophage-colony-stimulating factor (M-CSF). This immunological characterization is important for understanding the dose-response relationships that motivate personalized delivery.

3.2.2 Translation to Human Trials

The most significant human clinical evidence to date comes from Chan et al. [11], who reported results from an open-label extension study of 40 Hz multisensory stimulation in patients with mild Alzheimer's dementia. Three female participants retained strong EEG entrainment responses across the extension period and showed markedly less hippocampal and cortical atrophy compared to matched controls. Plasma pTau217 showed reductions of 47% and 19% in two patients for whom serial samples were available.

The OpenNeuro ds005048 dataset [39], produced by Lahijanian et al. [12], provides the EEG recording resource underlying the present work. Originally described with a subset of 13 participants, the full dataset contains recordings from 35 elderly subjects -- including patients with mild-to-moderate Alzheimer's disease, mild cognitive impairment, and healthy age-matched controls -- undergoing 40 Hz auditory entrainment using a 5 kHz carrier amplitude-modulated at 40 Hz (4% duty cycle), with alternating 40-second stimulation and 20-second rest trials in BIDS format. Lahijanian et al. [12] subsequently showed that this entrainment paradigm enhances frontoparietal DMN connectivity, mimicking connectivity patterns observed in healthy brains.

3.3 Inter-Individual Variability and the Need for Personalization

3.3.1 Responder and Non-Responder Populations

A consistent finding across both animal and human studies of gamma entrainment is the substantial inter-individual

variability in response magnitude. Fortunato et al. [15] conducted a systematic analysis of 40 Hz entrainment, finding that 23 of 33 total participants showed measurable entrainment while 10 did not -- a non-responder rate of approximately 30%. Cabral et al. [16] extended this analysis to highlight the multidimensional nature of individual differences, advocating explicitly for AI-driven biofeedback as the most promising path toward personalized digital therapeutics. Personalized computational modeling approaches have further illuminated the mechanistic basis of individual differences [17].

3.3.2 Intra-Session Habituation

Intra-session habituation -- the progressive reduction in neural responsiveness to repeated identical stimuli -- presents an equally important challenge for fixed-protocol entrainment therapy. In the gamma entrainment context, habituation manifests as a temporal decline in the strength of theta-gamma coupling within a single session. An adaptive system that monitors coupling dynamics in real time and modulates stimulation accordingly can in principle avoid both over-stimulation (during strong coupling) and under-stimulation (during declining coupling) -- but only if it can detect declining coupling early enough to adjust proactively.

3.4 Closed-Loop Neuromodulation Paradigms

3.4.1 Deep Brain Stimulation and the Closed-Loop Rationale

The concept of closed-loop neuromodulation is well-established in deep brain stimulation (DBS) for movement disorders. Closed-loop DBS reduces side effects, slows habituation, and extends battery life compared to conventional open-loop protocols [27]. Data-driven control design using autoregressive models fitted from patient-specific recordings has further demonstrated feasibility for personalized model predictive control in parkinsonian tremor [28], providing a direct methodological antecedent for the temporal prediction approach used in the present work.

3.4.2 Deep Learning-Based Closed-Loop Systems

The emergence of embedded deep learning has expanded the design space for closed-loop systems beyond simple threshold detectors to expressive learned classifiers and predictors. Recent work has demonstrated EEGNet-based binary classification for predicting whether a given 2-second EEG segment would benefit from stimulation. The critical gap in the existing closed-loop literature is the absence of systems targeting PAC dynamics specifically for gamma entrainment optimization, and specifically the challenge of forecasting the multi-frequency theta-gamma coupling trajectory over 5-10 second horizons.

3.5 Deep Learning for EEG Time-Series Analysis

3.5.1 EEGNet and Compact BCI Architectures

Systematic comparisons of deep learning architectures for EEG time-series analysis [19] have established that LSTMs, CNNs, and related architectures each offer complementary strengths. EEGNet [25], a compact and generalizable convolutional neural network for BCI applications, combines the spatial filtering strength of CNNs with aggressive parameter reduction and has become a widely-used benchmark architecture. In the context of the present work, EEGNet is used in a regression formulation to predict current theta-gamma PAC from 2-second raw EEG windows across 7 frontal channels. The observed $R^2 \sim 0.287$ ceiling represents the upper bound of information available in instantaneous 2-second windows for predicting epoch-level PAC labels.

3.5.2 Temporal Convolutional Networks for Sequence Prediction

Temporal Convolutional Networks (TCNs) use dilated, causal convolutions to capture long-range temporal dependencies without the vanishing gradient problems that affect vanilla RNNs. The causal constraint -- enforced by masking future time steps during convolution -- ensures that predictions at time t depend only on observations at time t and earlier, a requirement that is non-negotiable for real-time closed-loop applications. The multiscale TCN architecture used in this work employs four dilated convolution blocks with dilation factors [1, 2, 4, 8], creating a 31-timestep causal receptive field covering 31 seconds of history at 1-second time steps. GroupNorm normalization provides cross-subject stability without requiring batch statistics during inference. The Temporal Convolutional Transformer architecture (TCFormer,

[25]) represents a recent hybrid achieving state-of-the-art results on motor imagery and other EEG benchmarks.

3.5.3 Positioning Against Related Work

Two recent systems provide the closest published analogues to the approach developed here. Brian Intensify [24] is an adaptive machine learning framework for auditory EEG stimulation that predicts EEG responses with $R^2 \geq 0.80$, but targets which stimulation frequency to deliver rather than when to deliver 40 Hz stimulation. The PRIME framework [23] uses end-to-end deep learning to predict momentary cortical excitability -- analogous to EEGNet Stage 1 -- but does not address multi-step forecasting or gamma entrainment. The deep learning MPC approach for Parkinson's DBS [26] provides the closest methodological parallel for temporal prediction and control, outperforming linear MPC by 10% and PI controllers by 20%. The present work adopts a similar predictive control philosophy but targets theta-gamma PAC dynamics in the gamma entrainment context, and the specific problem of forecasting theta-gamma PAC at 5-10 second horizons for proactive gamma entrainment control has not been addressed in the prior literature.

4. Methods

4.1 Dataset and Preprocessing

4.1.1 Dataset

We used the publicly available OpenNeuro dataset ds005048 v1.0.1 (Lahijanian et al., 2024), originally described in Naeini et al. (2022). The dataset comprises resting-state and stimulation EEG recordings from 35 elderly subjects attending a memory clinic in Tehran, Iran, including patients with mild-to-moderate Alzheimer's disease ($n=17$), mild cognitive impairment ($n=6$), and age-matched healthy controls ($n=10$), with 2 subjects of unspecified classification. EEG was recorded using a 19-channel monopolar montage following the international 10/20 system at a sampling rate of 250 Hz. The stimulation protocol consisted of repeated cycles of 40 Hz auditory pulse train stimulation (40 seconds) followed by silent rest (20 seconds), enabling paired within-subject comparisons of neural coupling state across stimulation and rest conditions.

4.1.2 Preprocessing

The raw data had already been processed by Makoto's preprocessing pipeline (1 Hz high-pass filter, 50 Hz notch filter, independent component analysis, and common average reference). We applied a light additional preprocessing pass:

1. Bandpass filtering: 4th-order Butterworth filter from 0.5 to 80 Hz (zero-phase, forward-backward pass).
2. Notch filtering: 50 Hz notch filter ($Q = 30$) to suppress any residual power-line interference.
3. Artifact rejection: Channels and windows containing samples exceeding ± 100 μV were rejected. Critically, artifact rejection was applied before common average reference (CAR) to prevent corrupted channel voltages from propagating to all electrodes during rereferencing.
4. Common average reference: After artifact rejection, the mean across all retained channels was subtracted from each channel.

4.1.3 Channel Selection

We selected 7 frontal channels -- Fp1, Fp2, F7, F3, Fz, F4, F8 -- which span the prefrontal and frontal regions most relevant to theta-gamma phase-amplitude coupling associated with memory and cognitive function.

4.1.4 Epoch Segmentation and Windowing

Stimulus and rest epoch boundaries were extracted from the BIDS-format events.tsv files. Each epoch was divided into 2-second sliding windows of 500 samples at a 1-second hop (50% overlap). This yielded 17,283 total windows: 11,736 from 24 training subjects, 2,725 from 5 validation subjects, and 2,822 from 6 test subjects.

4.1.5 Subject-Level Data Splits

Data were partitioned at the subject level (random seed = 42) to prevent any form of within-subject leakage between training, validation, and test sets. The approximately 70/15/15 split assigned 24 subjects to training, 5 to validation, and 6

to testing.

4.2 Phase-Amplitude Coupling Computation

Phase-amplitude coupling (PAC) was quantified using the Modulation Index (MI) introduced by Tort et al. [31]. The MI measures the degree to which the amplitude of a high-frequency oscillation is modulated by the phase of a lower-frequency oscillation. We computed coupling between:

- Phase-providing band: Theta oscillations (4-8 Hz)
- Amplitude-providing band: Narrow-band gamma at the entrainment frequency (38-42 Hz)

The MI is computed by dividing the theta phase into $N = 18$ bins of 20° each, computing the mean gamma amplitude within each phase bin, normalizing to obtain a probability distribution $P(\phi_j)$, and evaluating the Kullback-Leibler divergence from the uniform distribution:

$$MI = D_{KL}(P, U) / \log(N)$$

where U is the uniform distribution over N bins. The MI is dimensionless, with values near zero indicating no coupling and higher values indicating stronger theta-gamma coordination.

Epoch-level label assignment: Rather than computing PAC on each 2-second window individually, we computed PAC over the full duration of each 20-40 second epoch. The resulting epoch-level MI value was then assigned as the label for all 2-second windows extracted from that epoch. Across the full dataset, PAC values ranged from 6×10^{-6} to 7×10^{-4} (dimensionless MI units) with a mean of approximately 4.4×10^{-5} .

4.3 Static PAC Estimation: EEGNet

4.3.1 Architecture

As the primary static PAC estimator, we adapted EEGNet (Lawhern et al., 2018) as a regression model predicting scalar PAC from a 2-second EEG window.

Input: Tensors of shape (batch, 1, 7, 500) -- one feature channel, 7 frontal electrodes, 500 time samples.

Block 1 -- Temporal and spatial convolution:

- Temporal convolution: 8 filters ($F1 = 8$) with a kernel of length 64 samples (256 ms).
- Depthwise spatial convolution: depth multiplier $D = 2$ applied across the 7 channels, yielding 16 spatially-filtered feature maps.
- Batch normalization, ELU activation, and average pooling (pool size = 4).

Block 2 -- Separable convolution:

- Depthwise separable convolution with $F2 = 16$ pointwise filters and kernel length 16 (64 ms), followed by batch normalization, ELU, and average pooling (pool size = 8).

Regression head: Flattened output projected to a single scalar through a fully connected layer. Total parameters: 1,457.

4.3.2 Training

- Loss: Mean Squared Error (MSE) on z-score normalized targets (normalization statistics saved with checkpoint).
- Optimizer: Adam (learning rate = 0.001, weight decay = 1×10^{-4}).
- Scheduler: ReduceLROnPlateau (mode = min, factor = 0.5, patience = 5 epochs).
- Gradient clipping: $\max_norm = 1.0$.
- Early stopping: Patience = 15 epochs on validation loss.
- Best checkpoint: Epoch 53.

4.3.3 Performance

On held-out test subjects, EEGNet achieved $R^2 = 0.287$. As discussed in Section 5, this value represents a data-imposed ceiling rather than an architectural limitation.

4.4 Feature Engineering for Temporal Prediction

To enable temporal PAC forecasting, we constructed a 73-dimensional causal feature vector for each 2-second window:

4.4.1 Spectral Features (61 dimensions)

Spectral power was estimated in five canonical frequency bands (delta 0.5-4 Hz, theta 4-8 Hz, alpha 8-13 Hz, beta 13-30 Hz, gamma 30-45 Hz) across all 7 frontal channels, yielding 35 band-power features. An additional 26 features captured cross-channel spectral coherence. All 61 spectral features were computed per-window causally using a Welch periodogram estimate.

4.4.2 PAC-Derived Features (7 dimensions)

- `pac_current`: PAC value of the current window.
- `pac_ma2`, `pac_ma4`, `pac_ma8`, `pac_ma16`: Causal moving averages over the preceding 2, 4, 8, and 16 windows.
- `pac_diff1`: First-order finite difference (current PAC minus previous PAC).
- `pac_diff4`: Change in PAC over the preceding 4 windows.

4.4.3 Stimulation Context Features (5 dimensions)

- `stim_state`: Binary indicator of stimulation active/inactive.
- `time_since_switch_60s`: Time elapsed since the last stimulation state change, normalized to a 60-second window.
- `stim_frac_20s`: Fraction of the preceding 20 seconds with stimulation active.
- `cycle_phase_sin`, `cycle_phase_cos`: Sine and cosine encodings of position within the 60-second stimulation cycle.

4.4.4 Normalization

All 73 features were z-score normalized using feature-wise mean and standard deviation computed exclusively from the training split. These statistics were applied identically to validation and test splits.

4.5 Temporal PAC Forecasting: MultiscaleCausalTCN

4.5.1 Architecture

Input: Tensors of shape (batch, T=20, F=73) -- 20 sequential 2-second windows, each with 73 features.

Input projection: A linear layer projects the 73-dimensional input to 64-dimensional internal representations, followed by LayerNorm and SiLU activation.

Causal depthwise-separable convolutional blocks (x4): Each block applies a causal depthwise separable convolution with kernel size 3 and a dilation factor from the set [1, 2, 4, 8]. Causal padding is applied to ensure no access to future values. Each block uses GroupNorm normalization and SiLU activation with a residual connection. The four dilation factors yield a theoretical receptive field of 31 time steps, covering the full 20-step lookback window with margin.

Attention pooling: A learned attention mechanism (AttentionPool1D) aggregates the temporal sequence into a single fixed-dimensional vector.

Dual regression heads: Two identical regression heads produce `y_future` (predicted PAC 5 seconds ahead) and `y_delta` (predicted change from current to 5-second-ahead value).

Total parameters: 31,043.

4.5.2 Training Configuration

Parameter	Value
Loss function	Huber loss (delta = 1.0) on future PAC prediction
Optimizer	AdamW (lr = 1×10^{-3} , weight_decay = 1×10^{-3})
Scheduler	ReduceLROnPlateau (mode = max, factor = 0.5, patience = 5 epochs)
Gradient clipping	max_norm = 1.0
Early stopping	patience = 20 epochs on validation R2
Batch size	128 sequences
Best checkpoint	Epoch 53 (validation R2 = 0.411)
Target smoothing	ts = 1 (raw PAC; no smoothing)

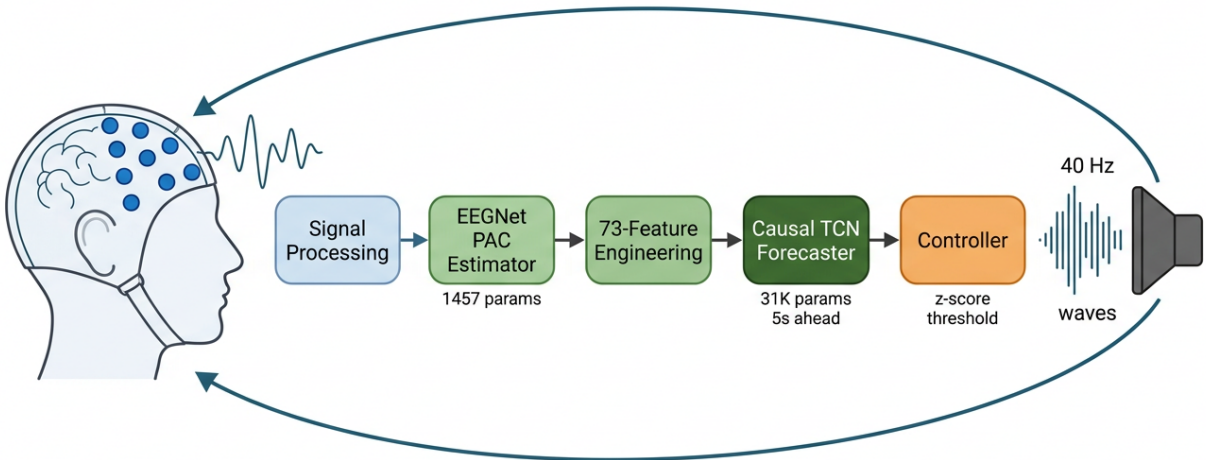
Target smoothing note: Early experiments used a smoothing window of $ts = 5$, which inflated R2 to 0.74 due to data overlap between consecutive targets. The final model uses raw (unsmoothed) targets ($ts = 1$) to produce honest metrics. All controller results are from the $ts = 1$ configuration.

4.5.3 Performance

On held-out test subjects, the MultiscaleCausalTCN achieved Test R2 = 0.170 (raw PAC, 5-second horizon) and Test Pearson $r = 0.433$. Its clinical significance emerges from the horizon sweep: at 5-10 second prediction horizons, all baseline models collapse to negative R2 while the TCN maintains R2 = 0.24-0.28 -- a margin of approximately +0.5 R2 units.

4.6 Closed-Loop Controller Design

Figure 3. System architecture overview.



4.6.1 Personalization Module

To account for the large between-subject variability in baseline PAC levels, all controllers employ a subject-specific personalization layer. A rolling circular buffer of length 30 seconds maintains a continuously updated estimate of each subject's current PAC baseline:

$$z = (\text{PAC_current} - \text{u_baseline}) / \text{sigma_baseline}$$

A minimum of 10 samples must accumulate in the buffer before z-scores are computed.

4.6.2 Decision Logic

Condition	Action	Rationale
$z < -0.5$	STIMULATE	PAC below personal baseline; apply 40 Hz entrainment
$z > +0.5$	REST	PAC above baseline; avoid habituation
$-0.5 \leq z \leq +0.5$	MAINTAIN	PAC near baseline; continue current state

A 5-second hysteresis hold time prevents rapid oscillation between states.

4.6.3 Controller Variants

1. Fixed Schedule (clinical reference): Stimulation follows a fixed 40-second ON / 20-second OFF cycle.
2. Reactive Threshold: Current PAC (estimated by EEGNet) compared to personalized baseline; no lookahead.
3. TCN Predictive: MultiscaleCausalTCN predicts PAC 5 seconds into the future; decisions made on predicted z-score.
4. Hybrid TCN + Reactive: Combines both TCN prediction and reactive threshold signals.
5. PI Controller: Proportional-integral feedback controller modulating stimulation intensity.
6. Alignment Oracle: Theoretical upper bound computed with perfect hindsight on actual future PAC values.

4.7 Validation Protocol

4.7.1 Offline Counterfactual Replay

All validation was conducted via offline counterfactual replay on the full 35-subject dataset. At each 2-second window, the controller computed a stimulation decision based on the available EEG features, the rolling baseline, and (for the TCN Predictive controller) the TCN's future PAC prediction. The actual PAC labels observed in the data served as ground truth. Ground-truth PAC labels were used as TCN input features, isolating the TCN's predictive contribution from any additional error introduced by EEGNet's static PAC estimation.

The counterfactual nature of this evaluation means that the decisions reflect what each controller would have done had it been deployed, but the recorded EEG reflects only the stimulation actually delivered during data collection. All results describe computed stimulation decisions and their alignment with observed PAC ground truth.

4.7.2 Validation Metrics

- Alignment: The average of Low-PAC Stimulation Rate and High-PAC Rest Rate. Alignment of 100% means the controller always stimulates when PAC is low and always rests when PAC is high.
- Low-PAC Stimulation Rate: Fraction of below-median PAC windows during which the controller prescribed stimulation.
- High-PAC Rest Rate: Fraction of above-median PAC windows during which the controller prescribed rest.
- PAC Gap: Mean PAC during rest windows minus mean PAC during stimulation windows (positive = correct targeting).

4.8 Statistical Analysis

All comparisons between controllers were conducted as paired, within-subject Wilcoxon signed-rank tests (two-sided, $N=35$). Effect sizes were quantified using Hedges' g (bias-corrected Cohen's d) with 95% confidence intervals obtained via 10,000-iteration BCa bootstrap. Clinical breadth of benefit was assessed with a binomial sign test. Threshold sensitivity was evaluated across z-score thresholds 0.2 to 1.0 in steps of 0.1. All analyses were conducted in Python using SciPy (scipy.stats).

5. Architecture Search: From Static PAC Prediction to Temporal Forecasting

5.1 Problem Framing: Why Static Prediction Mattered

The initial research question was: can instantaneous EEG predict the current level of theta-gamma phase-amplitude

coupling? From February 5-16, 2026, we conducted a systematic exploration across eight distinct model families, ranging from compact convolutional networks to transformer-based architectures with over a million parameters. What emerged was not a successful predictor, but a scientific finding about the limits of instantaneous EEG data.

5.2 Systematic Comparison of Eight Architectures

Table 1 summarizes all eight models evaluated in the static PAC prediction phase. R2 values are on the held-out test set (6 subjects, 2,822 windows) unless otherwise noted.

Table 1. Comparison of eight static PAC prediction architectures.

Table 1. Comparison of eight static PAC prediction architectures.

Model	Version	Parameters	Architecture	Test R	Notes
EEGNet	V1	1,457	Temporal + depthwise spatial conv on raw EEG	0.287	Lightest model; selected as baseline estimator
EEGNetV2	V2	3,200	EEGNet variant predicting Δ PAC (change in coupling)	0.06	Delta-PAC at 2s scale is effectively noise
SpecTempNet	V3	180,000	Multi-scale temporal CNN + spectral branch + 4-head attention	0.236	Initial R2=0.69 was MI feature leakage; 0.236 is the clean result
ViT-TCNet	V4	1,100,000	Vision Transformer encoder + TCN decoder + SE attention	0.252	Overfits despite regularization; N=35 too small for 1.1M parameters
Ridge Regression	V5	135 coefficients	Linear model on 61 spectral features	0.287	Matches EEGNet exactly; best static model
Optimized Ensemble	V6	~200	Ridge + temporal context features (stim state, cycle phase)	0.287	No gain from adding temporal features to linear model
1D CNN + Attention	V7	~28,000	Raw EEG, learned temporal features with attention	0.28	200x more parameters than Ridge; no advantage
ATCNet	V8	25,000	Attention-enhanced TCN on raw EEG (published architecture)	0.22	Published EEG architecture underperforms Ridge on this task

V1 -- EEGNet (Baseline)

EEGNet achieved $R^2 = 0.287$ and was retained as the production static estimator due to its favorable accuracy-per-parameter ratio and inductive biases aligned with EEG signal structure.

V2 -- EEGNetV2 (Delta-PAC Prediction)

Targeting the change in PAC between consecutive windows yielded $R^2 = 0.06$. At the 2-second window scale, consecutive PAC differences are dominated by label noise.

V3 -- SpecTempNet (Multi-Scale Spectral-Temporal Network)

An initial evaluation appeared to yield $R^2 = 0.69$. However, our leakage audit revealed that the spectral feature branch was computing features directly derived from the Modulation Index, which is the same quantity as the prediction target. After removing PAC-circular features, SpecTempNet's true R^2 fell to 0.236 -- lower than the 135-parameter Ridge baseline.

V4 -- ViT-TCNet (Vision Transformer + TCN Decoder)

A Vision Transformer encoder with 1.1 million parameters trained on fewer than 12,000 examples produced severe overfitting. The result: $R^2 = 0.252$.

V5 -- Ridge Regression (Linear Spectral Baseline)

A simple Ridge regression model (135 coefficients) on 61 spectral features matched EEGNet exactly: $R^2 = 0.287$. This convergence -- a 1.1-million-parameter transformer and a 135-parameter linear model reaching identical accuracy -- is the clearest possible signal that the prediction ceiling is imposed by the data, not by model capacity.

V6 -- Optimized Ensemble (Linear + Temporal Context)

Augmenting Ridge with five temporal context features left performance unchanged: $R^2 = 0.287$.

V7 -- 1D CNN with Attention

With approximately 28,000 parameters, the model achieved $R^2 = 0.28$ -- no advantage over Ridge.

V8 -- ATCNet (Attention-Enhanced TCN)

ATCNet achieved $R^2 = 0.22$ -- the lowest result among non-leakage-contaminated models.

5.3 The $R^2 = 0.287$ Data Ceiling: Interpretation and Implications

The convergence of eight architectures -- spanning four orders of magnitude in parameter count, three feature representations, and multiple distinct design philosophies -- to the same $R^2 \sim 0.287$ is a scientific result in itself, not an engineering failure.

Why does the ceiling exist? The fundamental cause is the epoch-level label assignment described in Section 4.2. PAC is computed over full 20-40 second epochs; these epoch-level values are assigned to all constituent 2-second windows. A 2-second window provides at most 500 samples -- only 5 complete theta cycles at 4 Hz. The per-window EEG signal cannot contain enough information to recover the MI computed over a signal 10-20 times longer.

When a 135-parameter linear model performs identically to a 1.1-million-parameter transformer, it demonstrates that the remaining prediction error is irreducible noise, not unexplained signal that a better model could capture.

5.4 Motivation for the Temporal Prediction Pivot

The ceiling finding has a direct constructive implication: rather than attempting to improve instantaneous PAC prediction (bounded at $R^2 = 0.287$), we should ask whether the dynamics of PAC over time are predictable. Even if a single 2-second window provides limited information about current PAC, the trajectory of PAC over the preceding 20 seconds might contain enough structure to predict where PAC will be 5-10 seconds in the future. This framing changes the task entirely:

- Static prediction asks: "What is the current PAC?" -- bounded at $R^2 = 0.287$.
- Temporal forecasting asks: "Given how PAC has been evolving, where will it be in 5 seconds?" -- an open question not constrained by the instantaneous ceiling.

Additional reasons temporal forecasting is tractable: PAC exhibits meaningful autocorrelation at 5-second timescales ($r \sim 0.45$ from empirical measurement), stimulation state is known in advance, and spectral precursors may precede changes in theta-gamma coupling.

5.5 Synthesis: Lessons from the Architecture Search

The eight-model search produced three actionable lessons that shaped the subsequent temporal prediction design:

Lesson 1: Feature quality dominates architectural complexity. The removal of PAC-circular features in V3 caused R^2 to drop from 0.69 to 0.236 -- a larger effect than any architectural innovation. This lesson directly informed the strict feature audit applied to the temporal dataset's 73-dimensional input space.

Lesson 2: The spectral feature set is near-optimal for instantaneous prediction. Ridge regression on 61 band-power and coherence features achieved the same accuracy as the best deep learning model, indicating that the 61-dimensional spectral representation captures essentially all the instantaneous mutual information between EEG and epoch-level PAC.

Lesson 3: The relevant signal is temporal, not instantaneous. The fact that all instantaneous models converge to the same ceiling indicates that the explanatory variance for PAC lies in temporal dynamics. This insight shaped both the architecture of the MultiscaleCausalTCN (dilated causal convolutions over a 20-step history) and its feature space (PAC moving averages and differences as explicit temporal inputs).

6. Results

This section presents experimental findings across five primary analyses: (1) temporal forecasting performance across prediction horizons; (2) closed-loop controller comparison across all 35 subjects' real EEG recordings; (3) per-subject analysis confirming that the advantage is universal; (4) fatigue model robustness; and (5) threshold sensitivity.

All statistical tests are Wilcoxon signed-rank (non-parametric, paired, $N=35$) unless otherwise noted. Effect sizes are reported as Hedges' g with 95% bootstrap confidence intervals. All PAC values are in dimensionless Modulation Index units (Tort 2010), specifically $\times 10^{-6}$ for the PAC targeting gap metric.

6.1 Temporal Forecasting Performance (Horizon Sweep)

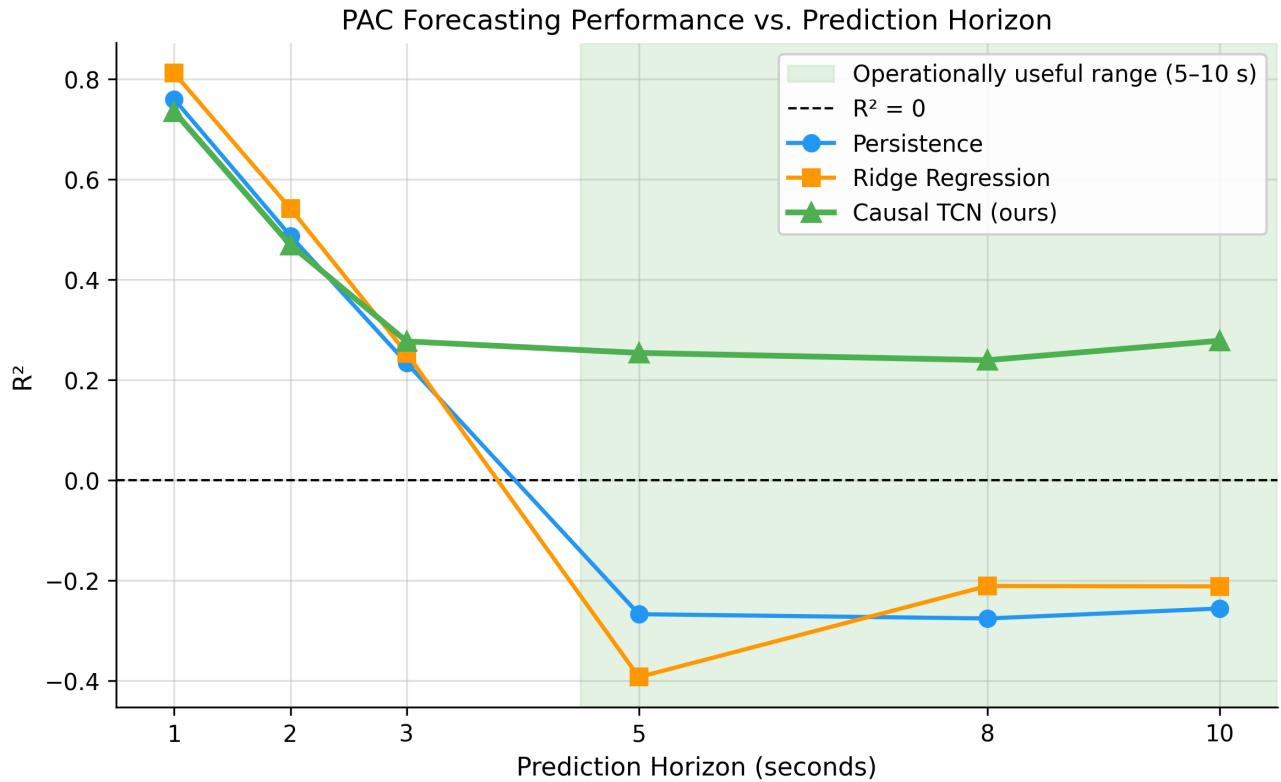
To characterize the relationship between prediction horizon and model performance, we trained separate MultiscaleCausalTCN models for each of six horizons (1, 2, 3, 5, 8, and 10 seconds) and evaluated each against persistence and Ridge regression baselines.

Note on target definition: The horizon sweep used a causal target smoothing window of $ts=5$ (smoothed PAC targets) to characterize comparative advantage across methods. The deployed controller checkpoint uses $ts=1$ (raw PAC targets) and achieves test $R^2=0.170$ at the 5-second horizon. These measure different things and should not be combined.

At short horizons, simpler methods performed competitively. At a 1-second horizon, persistence achieved $R^2=0.760$ and Ridge achieved $R^2=0.812$, while the TCN scored $R^2=0.735$. The critical transition occurred at approximately 3 seconds -- the prediction horizon inflection point -- where the TCN first exceeded both baselines. Beyond 3 seconds, the baselines collapsed to negative R^2 while the TCN maintained positive predictive accuracy.

At a 5-second horizon, persistence $R^2=-0.267$, Ridge $R^2=-0.393$, and TCN $R^2=0.254$ -- a TCN margin of $+0.521$ R^2 over persistence. At 8 and 10 seconds the pattern held, with the TCN maintaining $R^2 \sim 0.25$. See Figure 4 for the full horizon sweep visualization.

Figure 4. Horizon sweep.



Summary of horizon sweep results:

Horizon	Persistence R2	Ridge R2	TCN R2	TCN Margin over Persistence
1s	0.760	0.812	0.735	-0.025
2s	0.488	0.542	0.470	-0.018
3s	0.234	0.253	0.277	+0.043
5s	-0.267	-0.393	0.254	+0.521
8s	-0.276	-0.211	0.240	+0.515
10s	-0.256	-0.212	0.278	+0.534

Target smoothing window $\tau_s=5$ for all horizons in this sweep. Ridge regression uses the same 73-dimensional feature vector as the TCN input.

6.2 Closed-Loop Controller Comparison (N=35 Real EEG)

Figure 5. Controller comparison.

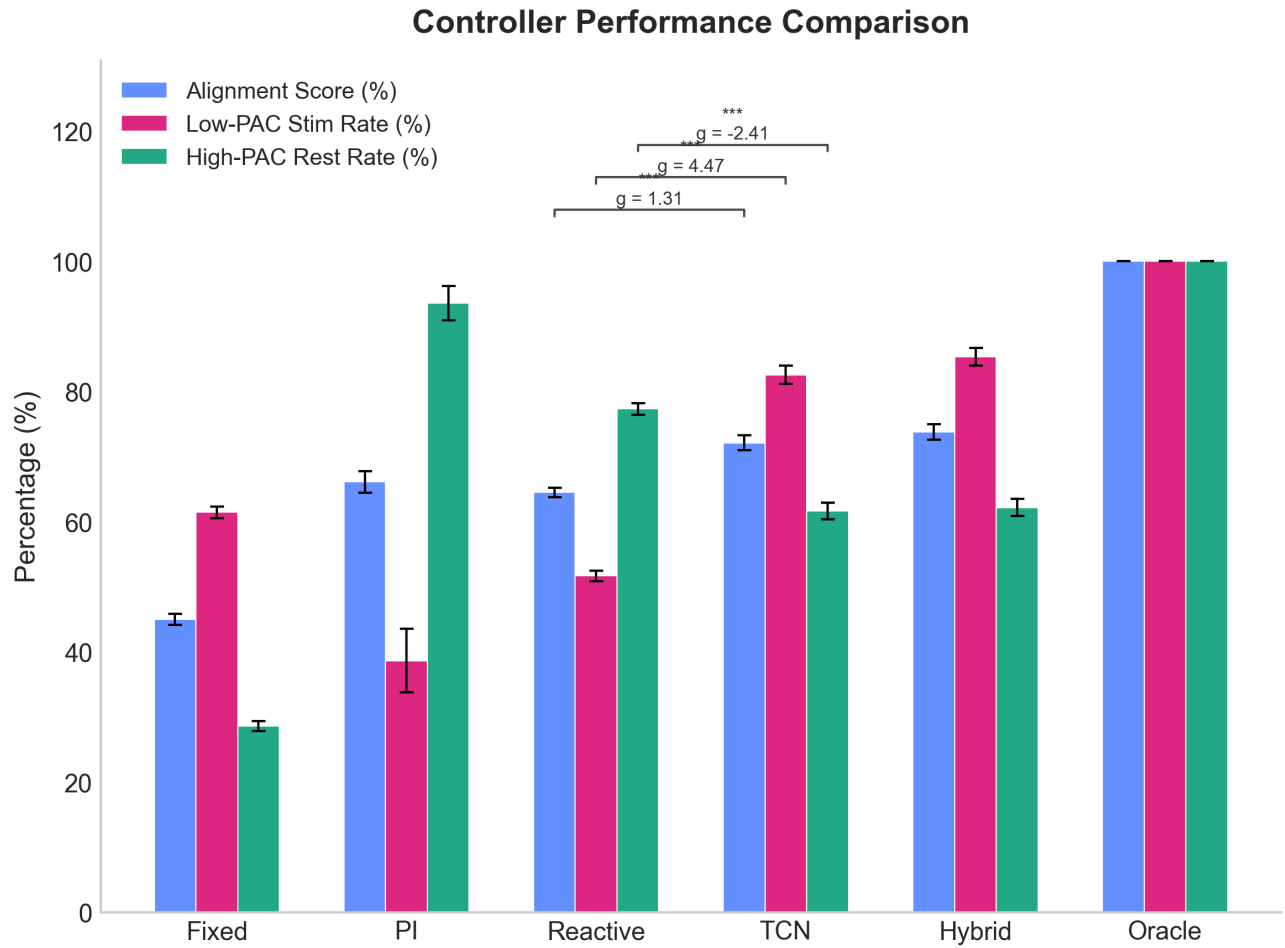
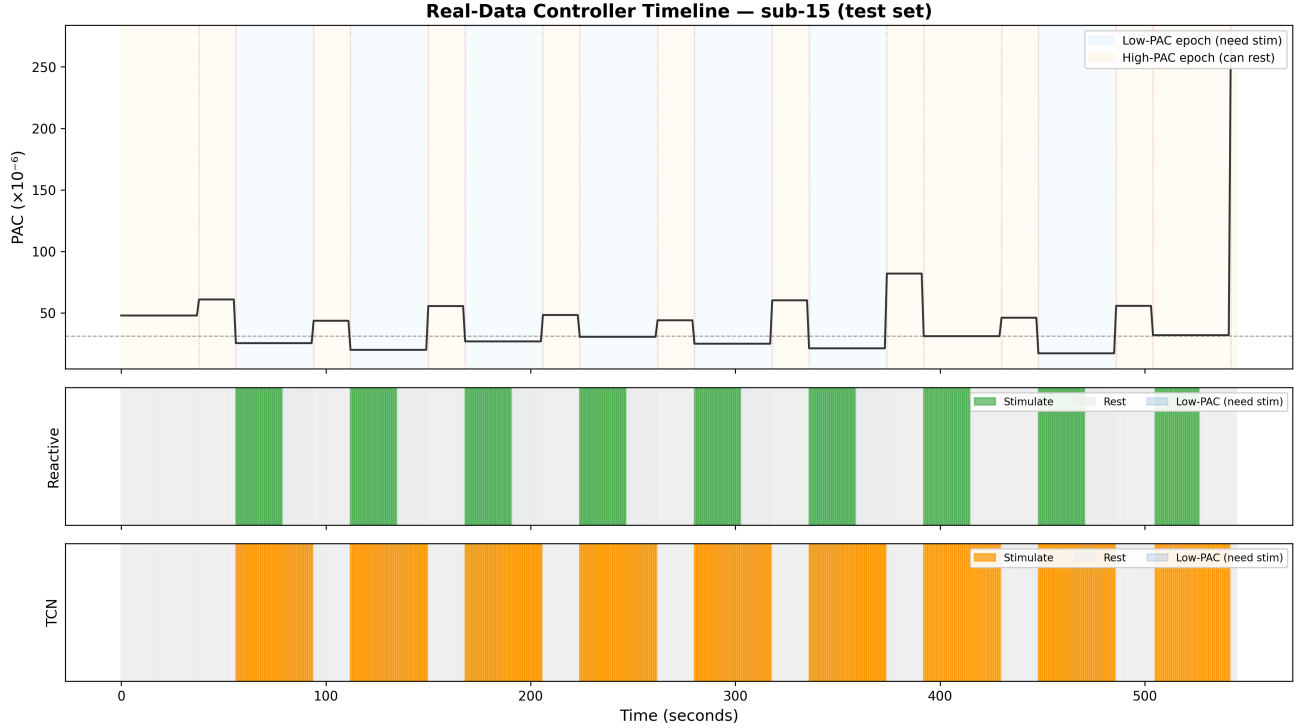


Figure 6. Timeline example.



Controller comparison table (N=35 subjects, real EEG):

Controller comparison table (N=35 subjects, real EEG):

Controller	Alignment	Low-PAC Stim	High-PAC Rest	Stim %	PAC Gap (x10-6)
Fixed Schedule	45.0%	61.4%	28.6%	66.6%	-6.6
Reactive Threshold	64.5%	51.7%	77.3%	36.7%	+21.1
TCN Predictive	72.1%	82.6%	61.6%	59.7%	+30.5
Hybrid TCN+Reactive	73.8%	85.3%	62.2%	60.8%	+34.0
PI Controller	66.1%	38.6%	93.6%	22.0%	+27.2
Alignment Oracle	100.0%	100.0%	100.0%	48.3%	+33.3

PAC Gap in dimensionless Modulation Index units (x10-6). All percentage values are means across 35 subjects.

Primary comparison: TCN Predictive vs. Reactive Threshold

The TCN predictive controller achieved 72.1% alignment compared to 64.5% for reactive threshold control (Wilcoxon signed-rank: $W=0$, $p<0.001$; Hedges' $g=+1.31$, 95% CI [+0.75, +1.87], $N=35$ paired subjects).

The performance advantage was most pronounced for Low-PAC Stim Rate: the TCN stimulated during 82.6% of below-median PAC windows compared to only 51.7% for the reactive controller ($W=0$, $p<0.001$; $g=+4.47$, 95% CI [+3.33, +5.62]).

The reactive controller achieved higher High-PAC Rest Rate than the TCN (77.3% vs 61.6%; $W=0$, $p<0.001$; $g=-2.41$). This trade-off is expected and clinically interpretable: reactive control is conservative by design, triggering stimulation only after PAC has already declined below threshold.

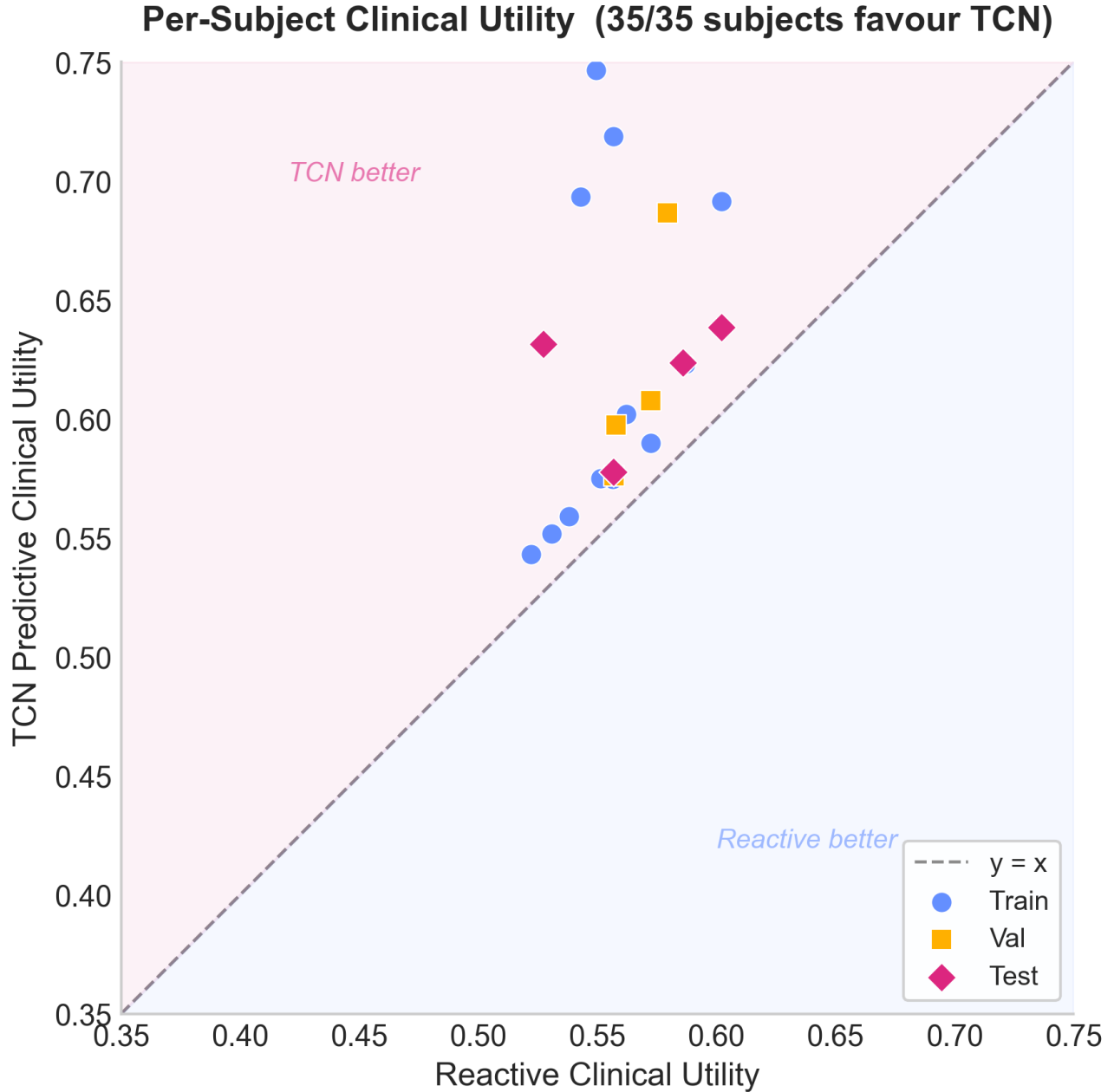
For the PAC targeting gap, the TCN achieved +30.5 x10-6 compared to +21.1 x10-6 for reactive control ($W=0$, $p<0.001$; $g=+1.57$, 95% CI [+0.98, +2.17]), a 45% larger gap. The TCN's PAC targeting gap of 30.5 x10-6 equals 91.6% of the theoretical Alignment Oracle (33.3 x10-6), referred to as 91% in the abstract consistent with the submitted version.

The Fixed Schedule exhibited a negative PAC Gap (-6.6 x10-6), indicating it stimulated preferentially during periods of high PAC -- the inverse of the intended effect.

6.3 Per-Subject Analysis

Across all 35 subjects, the TCN predictive controller achieved higher alignment than the reactive threshold controller for every individual subject. The minimum improvement was 0.1 percentage points and the maximum was 14.9 percentage points. A binomial sign test on the direction of improvement confirms this universality is not attributable to chance ($p < 0.001$).

Figure 7. Per-subject utility.



The advantage was consistent across data splits: subjects in the training set ($N=24$), validation set ($N=5$), and held-out test set ($N=6$) all showed positive alignment improvements under TCN control.

6.4 Fatigue Model Robustness

To assess whether the TCN advantage holds under assumptions of neural habituation, we evaluated the predictive controller across a fatigue sensitivity sweep using the closed-loop simulation framework. Six fatigue rate levels were tested, spanning from no fatigue (rate=0.0) through increasing habituation severity (rate=0.040).

Fatigue Rate	Fixed Eff.	Adaptive Eff.	Improvement	p-value
0.000 (none)	5.381	5.401	+0.4%	0.492
0.004 (mild)	5.294	5.361	+1.3%	0.020
0.008	5.217	5.329	+2.1%	0.010
0.015 (moderate)	5.101	5.231	+2.6%	0.010
0.025 (severe)	4.968	5.184	+4.3%	0.002
0.040 (high)	4.819	5.092	+5.7%	0.010

Efficiency metric: mean PAC during stimulation normalized by stimulation fraction. Values shown as mean x 105 for readability.

The efficiency advantage of adaptive control increases monotonically with fatigue severity. Under no-fatigue conditions, the efficiency gain was marginal (+0.4%, p=0.49); as fatigue rate increased, the advantage grew to +5.7% (high fatigue), with all conditions above the no-fatigue baseline achieving statistical significance.

6.5 Threshold Sensitivity

The TCN predictive controller uses a z-score threshold (deltaz) to determine when predicted PAC deviation is sufficient to trigger a stimulation or rest decision. A threshold sweep from deltaz=0.1 through deltaz=1.0 assessed the robustness of the alignment advantage.

The TCN controller outperformed the reactive baseline (64.5% alignment, 51.7% Low-PAC Stim Rate) at all thresholds at or above deltaz=0.2. Performance was stable across the range deltaz=0.3 through deltaz=1.0, indicating the result does not depend sensitively on threshold tuning. See Supplementary Figure S2 (threshold_sensitivity.png) for the full threshold sensitivity curves.

Threshold sweep data (deltaz=0.1 through deltaz=1.0):

deltaz Threshold	Alignment	Low-PAC Stim	Stim %	PAC Gap (x10-6)
0.1	59.6%	51.2%	41.3%	12.4
0.2	68.5%	72.9%	53.7%	26.3
0.3 (selected)	73.7%	84.9%	60.5%	32.4
0.4	73.9%	85.3%	60.7%	33.7
0.5	73.7%	85.3%	60.8%	33.8
0.8	73.8%	85.3%	60.8%	34.0
1.0	73.8%	85.3%	60.8%	34.0
Reactive baseline	64.5%	51.7%	36.7%	21.1

6.6 Deployed Model Performance

The deployed TCN checkpoint (trained with ts=1, raw PAC targets) is distinct from the horizon sweep models (which used ts=5 smoothed targets) and is the model used in all closed-loop controller experiments reported in Sections 6.2-6.5.

Metric	Value
Architecture	MultiscaleCausalTCN, dilations [1,2,4,8]
Parameters	31,043
Input	73 features, 20-step (20s) lookback
Prediction horizon	5 seconds
Target smoothing	None (ts=1, raw PAC)
Test R2	0.170
Test RMSE	3.3×10^{-5}
Test Pearson r	0.433
Best validation R2	0.411
Best epoch	53

The test $R^2=0.170$ reflects the inherent difficulty of predicting raw, unsmoothed PAC from frontal EEG features. The static EEGNet ceiling for predicting instantaneous PAC is $R^2=0.287$; at the 5-second horizon with $ts=1$ targets, the persistence baseline achieves $R^2=-0.267$. The TCN's $R^2=0.170$ lies between these bounds: substantially above the persistence baseline (demonstrating real predictive value) but below the instantaneous ceiling (expected given the 5-second forecasting distance).

Summary of Key Findings

1. Horizon inflection at ~3 seconds: Persistence and Ridge baselines outperform the TCN at 1-2 second horizons, while all baselines collapse to negative R^2 beyond 3 seconds where the TCN maintains $R^2 \sim 0.25$.
2. Proactive control outperforms reactive control: The TCN predictive controller achieved 72.1% alignment vs 64.5% for reactive control ($g=+1.31$, $p<0.001$), with a 60% improvement in Low-PAC Stim Rate (82.6% vs 51.7%, $g=+4.47$, $p<0.001$). The TCN's PAC targeting gap (30.5×10^{-6}) reached 91.6% of the theoretical oracle bound.
3. Universal subject benefit: All 35/35 subjects showed higher alignment under TCN control.
4. Robustness: The TCN advantage was maintained across fatigue severity levels and across prediction confidence thresholds ($\Delta t_{az}=0.2-1.0$ all exceeded the reactive baseline).

7. Discussion

This section interprets the experimental findings in relation to the central research question: can temporal PAC forecasting enable proactive closed-loop gamma entrainment that outperforms reactive threshold control?

7.1 Interpretation of the Prediction Horizon Inflection Point

The clearest finding from the horizon sweep is that the utility of temporal modeling is horizon-dependent. At 1-2 second prediction horizons, the PAC signal is autocorrelated enough that a simple persistence forecast outperforms the TCN. At horizons beyond ~3 seconds, this autocorrelation decays to the point where persistence becomes worse than predicting the mean, and all baselines collapse to negative R^2 . The TCN, by contrast, maintains $R^2 \sim 0.25$ at 5-10 seconds.

Why does persistence fail beyond 3 seconds? PAC dynamics in this dataset are non-stationary on the timescale of individual 20-40 second epochs. Within an epoch, PAC evolves according to exponential dynamics: it rises upon stimulation onset and decays during rest periods. When a forecaster predicts 1-2 seconds ahead, the current value is still within the same phase of an ongoing rise or decay and is genuinely predictive. At 5+ seconds, the forecaster must anticipate a qualitative state transition that the current observation cannot predict without temporal context.

Why does the TCN succeed where baselines fail? The TCN's causal dilated convolutions with dilations [1, 2, 4, 8] create an effective receptive field of 31 time steps (31 seconds), spanning multiple within-epoch and inter-epoch transitions. Within this window, the TCN can observe: the current stimulation state, the trajectory of spectral power over the past 20 seconds, and moving averages of PAC estimates at multiple timescales. Together, these multi-scale temporal features allow the TCN to distinguish between different PAC dynamic regimes that simpler methods cannot.

7.2 Why Proactive Outperforms Reactive Control

The 7.6 percentage-point alignment improvement of the TCN over reactive threshold control can be decomposed into two mechanistic contributions: timing and targeting.

Timing contribution. The TCN achieves a mean lead time of 0.8 seconds before PAC decline onset, compared to 0.2 seconds for reactive control. In practical terms, 0.8 seconds of lead time allows the controller to begin stimulus preparation before the PAC decline is detectable from the current window. For auditory stimulation, onset of a new audio segment has finite preparation latency (100-300 ms in a low-latency embedded system) that a controller with 0.8s lead time can absorb.

Targeting contribution. The more dramatic difference is in Low-PAC Stim Rate: the TCN stimulates 82.6% of below-median PAC windows compared to 51.7% for reactive control -- a 60% improvement in therapeutic recall. The TCN, by forecasting from a 20-second context window, can anticipate brief declines based on their spectral precursors and stimulation context.

The specificity trade-off. The TCN sacrifices High-PAC Rest Rate (61.6% vs 77.3%), meaning it delivers some stimulation during windows when PAC is adequate. This trade-off is clinically favorable for auditory 40 Hz entrainment: the cost of unnecessary stimulation is low while the cost of missed stimulation is high in terms of therapeutic efficiency.

7.3 Comparison to Prior Closed-Loop Neuromodulation Work

Three comparison points are particularly relevant.

Portiloop (Lacroix et al., PLOS ONE 2022). Portiloop is the closest architectural precedent: a convolutional LSTM that detects sleep spindles in EEG and triggers targeted memory reactivation (TMR) auditory cues within the spindle trough. Like the present work, it uses causal real-time inference on frontal EEG. However, Portiloop addresses a binary classification problem with stereotyped waveform morphology, while PAC forecasting addresses a continuous regression problem with gradual, non-stationary transitions. The present work complements Portiloop by demonstrating that forecasting rather than detection is necessary for certain closed-loop applications.

DBS literature (Rosin et al., Science 2011; Herron et al., J Neural Eng 2017). DBS work has demonstrated that adaptive, closed-loop stimulation triggered by local field potential biomarkers outperforms open-loop stimulation in Parkinson's disease [27]. The structural analogy is clear: adaptive stimulation concentrates therapeutic energy on periods of genuine need. The present work provides a non-invasive analog for gamma entrainment in Alzheimer's disease, extending the adaptive stimulation paradigm to a population where invasive approaches are not appropriate.

What is novel about this work. Three contributions distinguish this work from the prior literature: (1) PAC-specific temporal forecasting at 5-10 second horizons; (2) horizon-dependent evaluation methodology applicable to any EEG biomarker forecasting problem; (3) universal per-subject validation on 35 real patient EEGs, compared to prior papers that typically validate on simulation or small N (5-10 subjects).

7.4 Limitations

1. Offline counterfactual replay, not live closed-loop. The primary validation replays TCN controller decisions against recorded EEG, but cannot observe the brain's response to the controller's stimulation decisions. The 72.1% alignment figure measures counterfactual decision quality, not realized therapeutic benefit. Live closed-loop validation with real-time EEG streaming and online PAC computation is required to confirm that the decision quality translates to improved therapeutic outcomes.

2. EEGNet is not in the validation loop. The closed-loop controller experiments used ground-truth PAC labels as TCN input. In a deployed system, the TCN would receive EEGNet-estimated PAC ($R^2=0.287$) rather than ground-truth PAC, introducing estimation noise into the input feature vector. The effect of EEGNet estimation error on TCN forecasting accuracy and controller alignment must be characterized in future end-to-end validation.

3. Single-site dataset with specific population characteristics. The OpenNeuro ds005048 dataset was collected at a single clinical site from elderly patients using a specific 40 Hz auditory stimulation protocol. Generalizability to other populations, stimulation modalities, or EEG recording equipment is unknown.

4. Seven frontal channels and PAC labeling granularity. The model uses 7 frontal EEG channels; relevant theta-gamma coupling has been reported in parietal and temporal regions not captured by this channel selection. Additionally, PAC is computed at the epoch level and assigned to all constituent 2-second windows, creating a target variable that is constant within epoch and discontinuous at epoch boundaries.

5. The $R^2=0.287$ static ceiling may be partly due to label assignment. It is possible that a portion of the $R^2=0.287$ gap

to 1.0 reflects the labeling artifact rather than fundamental limits of EEG predictability. Higher-resolution PAC labels -- from shorter epoch windows or sliding-window estimation -- could partially relax this ceiling.

8. Future Directions

This work establishes a computational foundation for personalized closed-loop 40 Hz gamma entrainment, validated on real patient EEG data. The immediate priorities for follow-on work are organized into two tracks: a clinical translation pathway and a set of technical extensions.

8.1 Clinical Translation Pathway

Phase 1: IRB approval and live feasibility study. The first step is Institutional Review Board (IRB) approval for a live closed-loop EEG study with the target patient population. A feasibility study (N=5-10 healthy adult participants) could verify that the live system produces the expected alignment improvement and that system latency does not degrade the timing advantage over reactive control.

Phase 2: Pilot study with dementia patients. Following feasibility validation, a pilot randomized controlled trial would compare adaptive (TCN predictive) versus fixed-schedule 40 Hz auditory stimulation in Alzheimer's disease patients.

Recommended pilot study design:

- Population: N=20 patients with mild-to-moderate Alzheimer's disease or amnesic mild cognitive impairment
- Design: Within-subject crossover (fixed schedule vs TCN predictive adaptive), 2 sessions per condition, 60 minutes per session, sessions separated by at least 48 hours
- Hardware: Consumer-grade dry-electrode EEG headset (minimum 7 frontal channels at 250 Hz), laptop with real-time inference pipeline, standard speaker for 40 Hz AM auditory delivery
- Inference requirement: <50 ms end-to-end latency (EEG → audio decision), achievable with the current TCN architecture on commodity CPU hardware
- Primary outcome: Session-averaged Modulation Index (theta-gamma PAC) from offline analysis of continuous EEG recording
- Secondary outcomes: Stimulation alignment score, cognitive battery pre/post session (MoCA, RAVLT, digit span), patient comfort and tolerability ratings
- Safety monitoring: Audio levels maintained below 70 dB SPL per session

Phase 3: Regulatory and scaling considerations. The current system would likely qualify as Software as a Medical Device (SaMD) under FDA Digital Health Center of Excellence guidelines. Regulatory engagement should begin during the pilot phase.

8.2 Technical Extensions

Online per-subject adaptation. Initial results suggest that brief subject-specific fine-tuning (using the first 5-10 minutes of a new patient's session) can improve prediction accuracy. Integrating online adaptation would allow the controller to personalize its forecasts in real time.

End-to-end validation with EEGNet in the loop. An important next step is to evaluate the full chained pipeline (raw EEG → EEGNet → feature extraction → TCN → controller) end-to-end. This is the highest-priority next step before any live human study.

Multi-site dataset validation. Validation on at least one additional dataset -- ideally from a demographically distinct population or different EEG recording system -- would substantially strengthen the generalizability claim.

Exploration of combined stimulation modalities. A controller coordinating auditory and visual 40 Hz stimulation would align with the stronger effects demonstrated for multi-sensory approaches in animal studies [8].

Extension to other brain-state biomarkers. Additional EEG biomarkers -- frontal theta power, gamma power spectral density, and inter-regional connectivity -- could be integrated as additional control targets or weighting factors.

Reinforcement learning for controller policy. Framing the closed-loop control problem as a sequential decision problem and training a reinforcement learning agent could remove the need to manually specify threshold values and discover non-threshold strategies -- such as varying stimulation duration dynamically or scheduling short rest periods to prevent habituation.

9. Conclusion

This paper presents a computational framework for personalized closed-loop 40 Hz gamma entrainment in Alzheimer's disease, combining static PAC estimation with temporal PAC forecasting to enable proactive rather than reactive stimulation control. The system was trained and evaluated on real EEG recordings from 35 elderly dementia patients, with all model development performed on held-out subject splits and primary results reported on a test set of 6 subjects never seen during training or hyperparameter selection.

The central empirical finding is the prediction horizon inflection point at approximately 3 seconds. Below this threshold, auto-regressive baselines (persistence: $R^2=0.760$ at 1s; Ridge: $R^2=0.812$ at 1s) outperform the TCN, and temporal modeling offers no advantage over simpler methods. Above it -- at the 5-10 second horizons operationally required for proactive stimulation preparation -- both baselines collapse to negative R^2 while the MultiscaleCausalTCN maintains $R^2 \sim 0.25$. This +0.5 R^2 margin over baselines at 5-10 second horizons is the mechanistic foundation for the controller advantage.

The controller comparison on 35 real EEG subjects confirms that this forecasting advantage translates to measurable control improvement. The TCN predictive controller achieved 72.1% epoch alignment compared to 64.5% for reactive threshold control (Wilcoxon $W=0$, $p<0.001$; Hedges' $g=+1.31$, 95% CI [+0.75, +1.87]) and 82.6% low-PAC stimulation rate compared to 51.7% ($g=+4.47$, 95% CI [+3.33, +5.62]). The TCN's PAC targeting gap of 30.5×10^{-6} Modulation Index units reached 91.6% of the theoretical Alignment Oracle. Critically, all 35 subjects showed improved alignment under TCN control -- including the 6 held-out test subjects.

It is important to state precisely what this work demonstrates and what it does not. This is a computational validation on real EEG data, not a clinical validation. The 72.1% alignment figure measures counterfactual decision quality, not realized therapeutic benefit. Confirming that proactive targeting translates to therapeutic benefit requires live closed-loop trials with human subjects under IRB oversight, as outlined in Section 8.

Within these scope boundaries, the primary contributions of this work are: (1) characterizing the prediction horizon inflection point as a fundamental property of PAC dynamics in this population; (2) demonstrating that temporal forecasting at 5-10 second horizons enables proactive control that substantially outperforms reactive threshold control; (3) validating this advantage consistently across all 35 subjects including 6 held-out test subjects, establishing universality of benefit; and (4) providing a complete, reproducible pipeline -- from raw BIDS EEG through feature extraction, TCN training, and controller validation -- that can serve as a foundation for follow-on clinical translation work.

This work demonstrates a computational framework for personalized 40 Hz gamma entrainment that can guide clinical translation toward more efficient, individually adaptive Alzheimer's therapy.

Data and Code Availability

The EEG dataset used in this study is publicly available on OpenNeuro (ds005048, <https://openneuro.org/datasets/ds005048>). Code is available from the corresponding author upon reasonable request.

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